Experiments

Stat 256, Causality

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Identification vs. Estimation

Goal: learn about an *unobservable, counterfactual-dependent* quantity of interest using *observed* data.

Two inferential hurdles:

1. Identification: With data on an entire population, can you recover your QoI?
2. Estimation: With finite data on a sample, how well can you learn it?

Identification precedes estimation.

For each identification strategy we discuss:

1. the identification assumption
2. information from outside the data supporting it
3. testable implications (limited)
4. estimators
5. sensitivity to violations

Classical Randomized Experiment

Setup:

- Units: $i = 1, \ldots, N$
- Treatment: $D_i \in \{0, 1\}$, randomly assigned
- Potential outcomes: Y_{0i}, Y_{1i}
- Observed outcome: $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$ (consistency)
- $N_1 = \sum_{i=1}^{N} D_i, \quad N_0 = N - N_1$

Random assignment can take several forms:

- Complete randomization: Exactly N_1 units treated (draw without replacement)
- Simple (Bernoulli) randomization: Each unit independently assigned with $\Pr(D_i = 1) = p$

Randomization = Ignorability

Random assignment of D_i guarantees:

$$\{Y_{1i},\ Y_{0i}\} \perp\!\!\!\perp D_i$$

That is: the potential outcomes a unit *would have* under treatment or control are statistically independent of which group they are assigned to.

This is the identification assumption for experiments. Let's see what it buys us.

Identification of ATE Under Experiments

Quantity of Interest:

$$\tau_{ATE} \equiv \mathbb{E}[Y_{1i} - Y_{0i}]$$

Identification: Using $(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i$,

$$\begin{aligned}
\mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}] &= \mathbb{E}[Y_{1i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0] \quad \text{(by ignorability)} \\
&= \mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] \quad \text{(by consistency)} \\
&= \text{DIM estimand}
\end{aligned}$$

Practice these two steps. They are the core of every ID argument we will see.

Identification of Full Distributions

With $(Y_1, Y_0) \perp\!\!\!\perp D$, we identify more than means.

Identification Result

Let $F_{Y_d}(y)$ denote the CDF of Y_d. Then

$$\begin{aligned} F_{Y_0}(y) &= \Pr(Y_0 \leq y) = \Pr(Y_0 \leq y \mid D = 0) = \Pr(Y \leq y \mid D = 0) \\ F_{Y_1}(y) &= \Pr(Y \leq y \mid D = 1) \end{aligned}$$

So the effect at any quantile $\theta \in [0, 1]$ is identified:

$$\alpha_\theta = Q_\theta(Y_1) - Q_\theta(Y_0) = Q_\theta(Y \mid D{=}1) - Q_\theta(Y \mid D{=}0)$$

Note: the *quantile of the effect* $Q_\theta(Y_1 - Y_0)$ is **not** identified without further assumptions — the difference of quantiles is not the quantile of the difference.

SATE and PATE

- **SATE**: $\tau_{SATE} = \frac{1}{N}\sum_{i=1}^{N}(Y_{1i} - Y_{0i})$
  - Inferences limited to units in the study
  - Resampling notion can vary:
    - *Design-based*: $\boldsymbol{X}$, POs fixed; only treatment assignment is re-randomized
    - *Model-based*: $\boldsymbol{X}$, D fixed; residual influences on Y are random
- **PATE**: $\tau_{PATE} = \mathbb{E}[Y_{1i} - Y_{0i}]$ over a superpopulation
  - Generalizes the above to also redraw units from a larger superpopulation/DGP
- If sampling is random from the population, $\mathbb{E}[\text{SATE}] = \text{PATE}$. We are often uninterested in distinguishing them.

Variance of the DIM: PATE

Start from the variance of the mean difference:

$$\mathbb{V}(\hat{\tau}) \;=\; \mathbb{V}(\bar{Y}_1) + \mathbb{V}(\bar{Y}_0) - 2\mathrm{Cov}(\bar{Y}_1, \bar{Y}_0)$$

For the PATE (superpopulation view): treated and control units are independent draws, so $\mathrm{Cov}(\bar{Y}_1, \bar{Y}_0) = 0$:

$$\mathbb{V}_{\text{Neyman}}(\hat{\tau}) = \frac{\sigma^2_{Y_1}}{N_1} + \frac{\sigma^2_{Y_0}}{N_0}$$

This is the **Neyman variance**.

Plug-in estimator: replace population variances with sample variances:

$$\widehat{SE}(\hat{\tau}) = \sqrt{\frac{S_1^2}{N_1} + \frac{S_0^2}{N_0}}, \qquad S_d^2 = \frac{1}{N_d - 1} \sum_{i:D_i=d} (Y_i - \bar{Y}_d)^2$$

Variance of the DIM: SATE

For the SATE (finite-population / design-based view): the potential outcomes (Y_{1i}, Y_{0i}) are fixed. Randomness comes only from treatment assignment.

Under complete randomization, the N_1 treated units are a sample *without replacement* from the N fixed units. Two consequences:

- $\mathbb{V}(\bar{Y}_d)$ is *smaller* than $\sigma^2_{Y_d}/N_d$ by a finite population correction factor $(1 - N_d/N)$
- $\mathrm{Cov}(\bar{Y}_1, \bar{Y}_0) \neq 0$ because the two groups are complementary (selecting one determines the other)

Working through the finite-population sampling algebra, these combine to give:

$$\mathbb{V}_{SATE}(\hat{\tau}) = \frac{\sigma^2_{Y_1}}{N_1} + \frac{\sigma^2_{Y_0}}{N_0} - \frac{\sigma^2_{\tau}}{N}$$

where $\sigma^2_{\tau} = \mathrm{Var}(\tau_i) \geq 0$.

So the Neyman variance is **conservative** for the SATE. The gap is σ^2_{τ}/N: zero if effects are constant, and shrinking with N regardless.

Regression to Estimate ATE

Estimator (Regression with Experiments)

$$
\begin{aligned}
Y_i &= D_i Y_{1i} + (1 - D_i) Y_{0i} \\
&= Y_{0i} + (Y_{1i} - Y_{0i}) D_i \\
&= \underbrace{\bar{Y}_0}_{\alpha} + \underbrace{(\bar{Y}_1 - \bar{Y}_0)}_{\hat{\tau}} D_i + \underbrace{\{(Y_{0i} - \bar{Y}_0) + D_i[(Y_{1i} - \bar{Y}_1) - (Y_{0i} - \bar{Y}_0)]\}}_{\varepsilon_i}
\end{aligned}
$$

- This does **not** assume constant effects: $\hat{\tau}$ averages over heterogeneous τ_i.
- The residual ε_i has different variances under $D = 1$ and $D = 0 \Rightarrow$ heteroskedasticity.
- We need a heteroskedasticity-robust SE.

The Sandwich Variance Estimator

For OLS with $\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$:

$$\begin{aligned}\widehat{\mathbb{V}}(\hat{\beta} \mid \boldsymbol{X}) &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \, \widehat{\mathrm{Var}(Y \mid \boldsymbol{X})} \, \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \\ &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\hat{\Omega}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\end{aligned}$$

where $\hat{\Omega} = \mathrm{diag}(\hat{\omega}_1, \ldots, \hat{\omega}_N)$ estimates the conditional variance of each Y_i.

The choice of $\hat{\omega}_i$ defines the variant:

HC0: $\hat{\omega}_i = \hat{\varepsilon}_i^2$

HC1: $\hat{\omega}_i = \frac{N}{N-p}\hat{\varepsilon}_i^2$ (global degrees-of-freedom correction)

HC2: $\hat{\omega}_i = \frac{\hat{\varepsilon}_i^2}{1-h_{ii}}$ (leverage correction, h_{ii} = ith diagonal of $\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$)

With HC0, the "meat" of the sandwich becomes:

$$\boldsymbol{X}'\hat{\Omega}\boldsymbol{X} = \sum_{i=1}^{N} \hat{\varepsilon}_i^2 \, \boldsymbol{x}_i\boldsymbol{x}_i'$$

where $\hat{\varepsilon}_i = Y_i - \boldsymbol{x}_i'\hat{\beta}$ and $\boldsymbol{x}_i$ is the ith row of $\boldsymbol{X}$.

Sandwich for the Experimental Regression

The $[2,2]$ element of $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\text{diag}(\hat{\varepsilon}_i^2)\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}$ is just

$$\widehat{\mathbb{V}}(\hat{\tau}) = (\tilde{D}'\tilde{D})^{-1}\tilde{D}'\text{diag}(\hat{\varepsilon}_i^2)\tilde{D}(\tilde{D}'\tilde{D})^{-1}$$

where $\tilde{D}$ is the vector $D_i - \bar{D}$. This sorts and splits into two pieces:

$$= \frac{\sum_{i:\, D_i=1} \hat{\varepsilon}_i^2\, \tilde{D}_i^2}{\left(\tilde{D}'\tilde{D}\right)^2} + \frac{\sum_{i:\, D_i=0} \hat{\varepsilon}_i^2\, \tilde{D}_i^2}{\left(\tilde{D}'\tilde{D}\right)^2}$$

Noting that $\tilde{D}_i = N_0/N$ for treated, and $-N_1/N$ for controls, and that $\tilde{D}'\tilde{D} = N_1 N_0/N$, after some tedious algebra we get this to equal

$$= \frac{\widehat{\text{Var}}(Y|D=1)}{N_1} + \frac{\widehat{\text{Var}}(Y|D=0)}{N_0}$$

Further, if you construct $\widehat{\text{Var}}(Y|D=d)$ with the $N_d - 1$ correction then you have the HC2 correction and this matches Neyman exactly.

Example: Effect of Job Training on Earnings

- Treatment: $N_1 = 7{,}487, \quad \bar{Y}_1 = \$16{,}199, \quad S_1 = \$17{,}038$
- Control: $N_0 = 3{,}717, \quad \bar{Y}_0 = \$15{,}040, \quad S_0 = \$16{,}180$
- $\hat{\tau} = 16{,}199 - 15{,}040 = \$1{,}159$
- $\widehat{SE} = \sqrt{\frac{17{,}038^2}{7{,}487} + \frac{16{,}180^2}{3{,}717}} \approx \$330$
- $t = 1{,}159 \,/\, 330 \approx 3.5, \quad p \approx 0.0005$

All exactly the same thing:

- DIM with Neyman SE
- Welch (unequal variance) t-test
- `lm(earnings ~ D)` with HC2 SEs via `sandwich::vcovHC(fit, type="HC2")`

Fisher's Exact Test

You already know the large-N test:

$$H_0 : \mathbb{E}[Y_1] = \mathbb{E}[Y_0], \quad H_1 : \mathbb{E}[Y_1] \neq \mathbb{E}[Y_0] \qquad \text{(weak null)}$$

Fisher's Exact Test uses a stronger hypothesis:

$$H_0 : Y_{1i} = Y_{0i} \;\; \forall\, i \qquad \text{(sharp null of no effect)}$$

Key idea: Under the sharp null, we "observe" all potential outcomes — so we can compute the test statistic under every possible treatment assignment.

Procedure:

1. Compute $\hat{\tau}_{\text{obs}}$ from actual assignment
2. For every (or many) alternative assignments $\omega \in \Omega$, compute $\hat{\tau}(\omega)$ using the filled-in PO schedule
3. p-value $= \Pr_\omega\big(|\hat{\tau}(\omega)| \geq |\hat{\tau}_{\text{obs}}|\big)$

Fisher's Exact Test: Small-Sample Example

i	Y_{1i}	Y_{0i}	D_i
1	3	?	1
2	1	?	1
3	?	0	0
4	?	1	0
$\hat{\tau}_{\text{obs}}$			1.5

What do we know given $H_0 : Y_{1i} = Y_{0i} \ \forall i$?

i	Y_{1i}	Y_{0i}	D_i
1	3	3	1
2	1	1	1
3	0	0	0
4	1	1	0
$\hat{\tau}_{\text{obs}}$			1.5

Sharp null fills in the full schedule. Now enumerate $\binom{4}{2} = 6$ assignments

Scaling Up

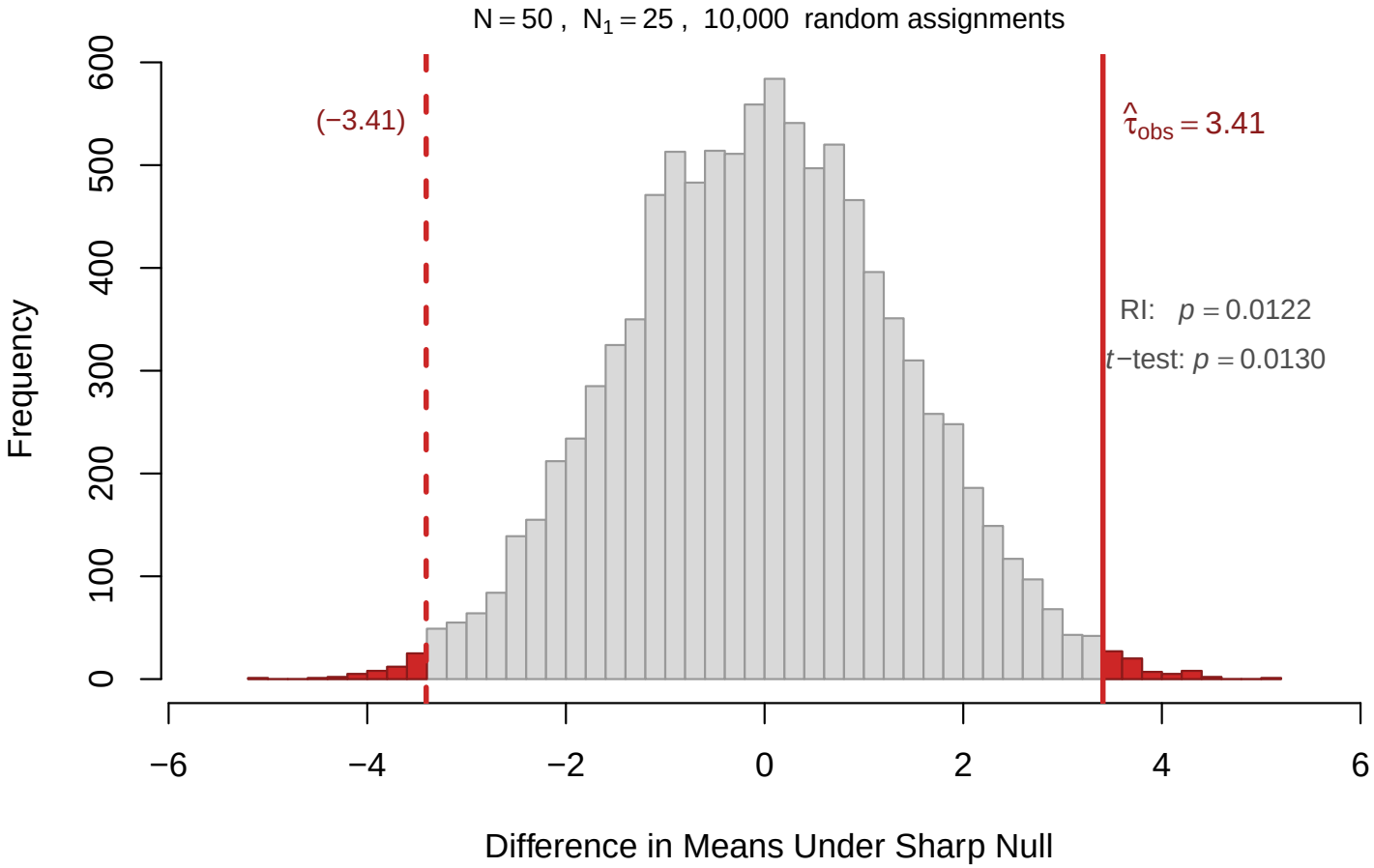
With 4 units and 2 treated: $\binom{4}{2} = 6$ assignments — enumerate all.

With 50 units and 25 treated: $\binom{50}{25} \approx 1.26 \times 10^{14}$.

Solution: sample from Ω.

- Re-draw treatment assignment many times (e.g. 10,000)
- Compute $\hat{\tau}(\omega)$ each time under the sharp null
- Compare observed $\hat{\tau}$ to this simulated null distribution

Randomization Inference in Practice



Histogram: 10,000 re-randomizations under the sharp null.
Solid line: observed $\hat{\tau}$. Shaded: values at least as extreme $\Rightarrow$ p-value.

How Do These Inferences Differ?

Finite-sample exactness.

- The RI p-value is exact under the sharp null — no CLT required.
- When N is large, the null distribution is approximately normal and you get nearly the same answer as the t-test.

Sharp null $\neq$ weak null, but close in practice.

- The sharp null says $Y_{1i} = Y_{0i}$ for *every* unit.
- But a summary statistic like the DIM is only sensitive to effects that *accumulate* — not to effects that cancel. Rejecting the sharp null via the DIM really tells you "effects exist *and* don't average to zero," which is close to the weak null.

RI vs. the bootstrap.

- (Paired) bootstrap resamples *units* — "what if we'd drawn different people?" (superpopulation).
- RI re-randomizes *treatment* within the fixed sample — "what if the same people got different assignments?" (design-based).

Bottom line: RI most valuable in small samples where CLT inference is suspect.

When Observations Are Not Independent

Everything so far assumed (conditionally) independent observations. Often violated:

- Voters in the same precinct
- Students in the same classroom
- Repeated measures on the same person

Two distinct problems:

1. **Clustered assignment.** Treatment assigned at group level (village, school, precinct). This inflates the *actual* variance of $\hat{\tau}$ — you have less independent information than N suggests.
2. **Correlated residuals.** Even with individual assignment, within-group correlation in ε_i makes standard SEs *too small* — the SE *estimate* is wrong, even if actual variance is fine.

The dangerous case is **both at once**: a cluster-level component in Y *and* no within-cluster treatment variation to cancel it out.

Cluster-Robust SEs

The cluster-robust ("CR") sandwich estimator:

$$\widehat{\mathbb{V}}_{CR}(\hat{\beta}) = (\boldsymbol{X}'\boldsymbol{X})^{-1} \left(\sum_{c=1}^{C} \boldsymbol{X}_c' \hat{\varepsilon}_c \, \hat{\varepsilon}_c' \boldsymbol{X}_c \right) (\boldsymbol{X}'\boldsymbol{X})^{-1}$$

where $\boldsymbol{X}_c$, $\hat{\varepsilon}_c$ are the rows and residuals for cluster c.

Allows **arbitrary** intra-cluster correlation; assumes clusters are independent.

Practical notes:

- Cluster on the source of dependence in your residuals, not on every grouping you can think of.
- With few clusters ($C \lesssim$ 30–50), CR-SEs are downward biased. Alternatives: wild cluster bootstrap (`fwildclusterboot` in R), randomization inference at the cluster level, or aggregate to cluster means.
- If there is *no* cluster-level component in Y, or if treatment varies within cluster so the component cancels, then HC0 is fine — CR-SEs buy nothing and widen CIs.

Unbiased Is Not Enough

Under random assignment, $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$ is unbiased for the ATE.

But "unbiased" is a statement about averages over hypothetical repetitions. In any *single* experiment, $\hat{\tau}$ can be far from τ.

What we care about is **root mean squared error**:

$$\text{RMSE}(\hat{\tau}) = \sqrt{\mathbb{V}(\hat{\tau}) + \text{Bias}^2} = \sqrt{\mathbb{V}(\hat{\tau})} \quad \text{(when unbiased)}$$

So the goal is to **reduce variance**. Two levers:

1. **Design**: blocking (build balance in before randomization)
2. **Analysis**: covariate adjustment (remove imbalance after the fact)

Both exploit the same idea: a pre-treatment covariate X that predicts Y can absorb residual variation.

Blocking

Idea: pre-stratify on a strong predictor of Y, then randomize treatment *within* each stratum (block).

- Guarantees exact balance on the blocking variable, in your sample — not just in expectation.
- Within-block variation in Y is smaller, so the within-block DIM is sharper.
- Randomization still balances unobserved influences, in expectation.

Estimator: within each block j, compute $\hat{\tau}_j$. Then weight by block size:

$$\hat{\tau}_B = \sum_{j=1}^{J} \frac{N_j}{N} \hat{\tau}_j$$

Equivalently (when treatment probability is the same in every block): OLS with block fixed effects.

Rule of thumb: block on the strongest predictor of Y you have. A pre-treatment version of the outcome is ideal. Blocking on something unrelated to Y costs nothing but gains nothing.

Covariate Adjustment: The Lin Estimator

Adjustment removes the contribution of chance covariate imbalances *after* they happen (in analysis, not design).

The simplest approach:

$$Y_i = \alpha + \tau D_i + \beta' X_i + \varepsilon_i$$

This is fine in large samples, but has a small $O(1/N)$ bias (Freedman 2008).

Lin (2013): demean X and interact with D:

$$Y_i = \alpha + \tau_{Lin} D_i + \beta_1'(X_i - \bar{X}) + \beta_2' D_i \cdot (X_i - \bar{X}) + \varepsilon_i$$

- τ_{Lin} is directly interpretable as the ATE.
- Avoids misspecification biases under heterogeneous effects that otherwise lead to "weird weights" and Freedman's $O(1/N)$ bias critique.
- Use HC2 SEs. Pre-specify the covariates.

Preview: this is nothing but *regression imputation* / *G-computation* / the *T-learner* — fit $\hat{\mu}_d(X)$ separately per arm, impute both POs, average. We will return to this.

Best practice: "Block on what you can, adjust for what you can't." The two

Takeaways

- **Randomization = ignorability.** Identification uses two steps: ignorability, then consistency.
- **DIM = Neyman = Welch t = OLS with HC2.**
- **Randomization inference** is exact in finite samples and design-based. Rejecting the sharp null via DIM is close to rejecting the weak null.
- **Clustering** inflates variance and/or breaks SE estimation. Cluster-robust SEs handle it with enough clusters.
- **Blocking and covariate adjustment** reduce variance — unbiasedness alone is not the goal. Block on what you can, adjust for what you can't.
- **Next**: What if we can't randomize?

Appendix

Why Neyman Is Conservative for SATE

Recall the exact variance:

$$\mathbb{V}(\hat{\tau}) = \frac{\sigma^2_{Y_1}}{N_1} + \frac{\sigma^2_{Y_0}}{N_0} - \frac{\sigma^2_\tau}{N}$$

The Neyman approximation drops $-\sigma^2_\tau/N$:

$$\mathbb{V}_{\text{Neyman}}(\hat{\tau}) = \frac{\sigma^2_{Y_1}}{N_1} + \frac{\sigma^2_{Y_0}}{N_0}$$

Since $\sigma^2_\tau \geq 0$, we have $\mathbb{V}_{\text{Neyman}} \geq \mathbb{V}(\hat{\tau})$.

How large is the gap?

- If effects are constant ($\tau_i = \tau$ for all i), then $\sigma^2_\tau = 0$ and Neyman is exact.
- If effects vary, Neyman overestimates. The gap is σ^2_τ/N, which shrinks with N.

Either way, CIs based on Neyman SEs have *at least* nominal coverage for the SATE.